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| <b>ACTIGAS/ACTIOPA/ACTIGO EXHAUST<br/>EMISSION TESTERS<br/>EXPORT VERSION</b> | 1/6        | <b>N° : F2-B</b> |
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**1/ COMMERCIAL REFERENCES**

| Référence | Désignation                                         |
|-----------|-----------------------------------------------------|
| ACTIGASxx | Gas analyser with black streamlined console         |
| ACTIOPAxx | Opacimeter with black streamlined console           |
| ACTIGOxx  | Gas-opacimeter combo with black streamlined console |
| OPA-7276  | opacimeter exhaust extension                        |
| AT1045007 | Sampling probe d10mm-l=3,5                          |
| ❖         | Xx indicates the country code                       |
| XP        | Standart Export (adaptive Program)                  |
| GR        | Greek                                               |
| MA        | Moroccan Arabic                                     |
| RU        | Russian                                             |
| SE        | Serbian                                             |

**2/ PRODUCT PICTURES**

Gas analyser



Gas-opacimeter combo



Exhaust extension (in option)

|                                                                               |            |                  |
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**3/ REFERENCE DOCUMENTS**

| Documents de références      | référence                        | Indice     | Langue |
|------------------------------|----------------------------------|------------|--------|
| Operating manual ACTIGAS     | ACTIGAS manual export version-en | 05/03/2012 | GB     |
| Operating manual ACTIOPA     | 605 user manual muller-en        | 10/03/2009 | GB     |
| ACTIGAS technical data sheet | DOC M2/170                       | 0          | FR/GB  |
| ACTIOPA technical data sheet | DOC M2/172                       | 0          | FR/GB  |
| ACTIGO technical data sheet  | DOC M2/171                       | 0          | FR/GB  |
| Gas analyser certificate     | SK 08 -001 MI-010 of 06/10/2010  | Add 3      | GB/CZ  |
|                              |                                  |            |        |
|                              |                                  |            |        |

**4/ AVAILABLE ELECTRICAL POWER SUPPLIES**

| Voltage / Frequency |          |
|---------------------|----------|
| 230 v single-phase  | 50/60 Hz |

**5/ EQUIPMENT DESCRIPTION**

Depending on the « xx » code used, the devices are delivered either with a standard export programme or with a specific programme.

At present, only the following versions are available :

| Country code | Software Version                   |
|--------------|------------------------------------|
| XP           | Standart Export (adaptive Program) |
| GR           | Greek                              |
| MA           | Moroccan Arabic                    |
| RU           | Russian                            |
| SE           | Serbian                            |

**5.1 MANUFACTURE**

The equipment is manufactured in the Czech Republic by ATAL, a subsidiary of ACTIA.  
The workstation includes a PC, to which the gas and/or opacimeter units are connected.  
The Gas unit is designed and manufactured by ATAL.  
The opacimeter unit is developed by SENSOR, model LCS2400  
The MMI software for the gas analyser and opacimeter is developed by ATAL.

**5.2 ETAT DE LIVRAISON**

- All three versions consist of a single console:
- In the case of a gas analyser, a gas unit is added, fitted to the console.
- For the opacimeter, the unit is packed separately. The recess for the gas unit is closed off with a hatch.
- For the gas-opacimeter combo, the above two configurations are combined, except that no hatch is added.

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### 5.3 LEGAL METROLOGY

- As regards legal metrology, this equipment is quite different to other exhaust emission testing equipment on the market.
- The PC, which is common to both applications, is no longer part of the approvals, unlike earlier CPUs, which were included in the primitive testing and were sealed.
- The gas analyser unit was approved in Slovakia, as per the European Measuring Instruments Directive (MID)
- The opacimeter cell was certified by the LNE (National laboratory of French Essays).

During installation, the technician must enter the serial numbers of the measuring cells into the exhaust emissions application software. This procedure, protected by a password, makes it impossible for anyone other than an approved technician to replace a measuring cell.

## 6/ TECHNICAL SPECIFICATIONS

### 6.1 ANALYSEUR DE GAZ

Gas analyser in compliance with European directive of Measuring instruments(MID)  
Equipment delivered as a castor-mounted workstation, test lane type. With its very compact design, it can accommodate all peripheral accessories.

The 19" swivelling flat screen makes it possible to monitor the testing operations from inside the vehicle.

Supplied with integrated A4 printer, oil temperature sensor, battery/cigarette lighter-type rev. counter, AZERTY keyboard, mouse, infra-red remote control

#### CHARACTERISTICS

- Mini industrial PC with DVD drive
- Measuring cell
- Carbon monoxide (CO) 0 to 10 % vol
- Carbon dioxide (CO<sub>2</sub>) 0 to 20 % vol
- Unburnt hydrocarbons (HC) -12 to 2000 2001 to 9000 ppm vol
- Oxygen (O<sub>2</sub>) 0 to 4 % 4 to 21 % vol
- Lambda ratio (λ) 0,500 to 2,000
- Engine speed 400 to 2000 rpm and 2010 to 9990 rpm
- Oil temperature 5 to 150 °C
- Nitrogen oxides (progressive) (NOX) 0 to 5000 ppm vol
- Pump capacity 10 l / min
- Barometric compensation 50 h Pa to 1060 h Pa
- Class 0 device,OIML R99/0 compatibility



OPTIONAL: 50 m RJ45 cable interfacing between the central PC and the Gieglan protocol



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## 6.2 OPACIMETER

Opacimeter compliant with standard NFR 10025

Equipment delivered as castor-mounted workstation, test lane type. With its very compact design, it can accommodate all peripheral accessories.

The 19" swivelling flat screen makes it possible to monitor the testing operations from inside the vehicle

Supplied with integrated A4 printer, oil temperature sensor, battery/cigarette lighter-type rev. counter, AZERTY keyboard, mouse, infra-red remote control

Tripod-mounted exhaust fumes measuring cell

### CHARACTERISTICS

- Mini industrial PC with DVD drive
- Opacity 0 to 99.0 %
- Opacity 0 to 9; 99 Km-1
- Measuring chamber length 200 mm
- Measuring pipe length 1200 mm
- Battery rev counter from 400 to 2000 and 2001 to 9990 rpm
- Engine oil temperature 5 to 150 °
- Measuring chamber temperature 70 to 95 °
- Exhaust fumes measurement 0 to 600 °
- Power supply: 220 V single-phase



OPTIONAL: 50 m RJ45 cable interfacing between the central PC and the Gieglan protocol





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### **6.3 COMBINED WORKSTATION**

The combined workstation features the same characteristics as both the devices described above

### **6.4 BENEFITS OF THE DEVICES**

- Large screen
- Ergonomic and user-friendly human-machine interface
- Self-repair, keyboard, mouse, screen, printer
- CPU runs with Windows XP
- Remote maintenance for remote troubleshooting
- Original GIEGLAN link
- Can be used as screen repeater for the test lane
- Measuring cells are easy to repair
- Modular device
- 

### **Standard Export Version :**

This is an adaptive programme. Significant changes to the procedure may be made throughout the product's service life :

- Redefinition of preparation and measuring ramp-ups
- Redefinition of new limit values
- Redefinition of display and print

### **7 / OPACIMETER EXHAUST EXTENSION**

Owing to the design of the exhaust system on certain vehicles, (non removable grille, exhaust outlet diameter too small, elbowed outlet), it is impossible to insert the opacimeter.

In this case, regulations require that an exhaust extension be used, into which the opacimeter probe is inserted.

- The extension is an optional element; it is not included in the composition.



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**7 / APPROVED SAMPLING PROBE FOR VERTICAL EXHAUST  
STACKS OF HEAVY VEHICLES**

This probe, measuring 3.5 m in length, is heated and approved for the technical inspection of heavy vehicles.

The probe is delivered with its power supply unit and handling boom.



**8/ SPECIAL TECHNICAL INFORMATION (in chronological order of publication)**

Not applicable

**HISTORIQUE DU BULLETIN DE VENTE**

| Modification                                                 | Indice du bulletin | Date       | Rédacteur         |
|--------------------------------------------------------------|--------------------|------------|-------------------|
| Creation of the bulletin                                     | -                  | 20/02/2012 | E.CASTREC / C.ROY |
| Bulletin updated ( addition of opacimeter exhaust extension) | A                  | 05/07/2012 | ROY               |
| Bulletin updated ( addition of sampling probe)               | B                  | 02/08/2012 | ROY               |
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